

Machine Learning Engineer Assessment

Freight Rate Prediction — Spotter

1. Objective

The objective was to train a model using the labeled development data, validate the modeling approach, generate predictions for all 12,000 validation loads, and produce the required December 2025 prediction chart.

2. Data Exploration and Quality

The development dataset contains 48,000 labeled loads covering January through October 2025. The target variable, `posted_rate`, is complete. Missing values were present in `market_index` (374 rows) and `weight` (300 rows). The target distribution is strongly right-skewed, and distance is a major driver of the freight rate.

3. Validation Strategy

I used chronological validation rather than a random split because the final prediction set is in later months than the labeled development data. Two forward-looking checks were used: training on January–August and validating on September; then training on January–September and validating on October.

Training period	Validation	MAE	RMSE	MAPE
Jan–Aug 2025	Sep 2025	17.17	186.22	0.47%
Jan–Sep 2025	Oct 2025	17.98	159.42	0.59%

These are internal validation measurements. The final Spotter validation metric is calculated after submission.

4. Feature Engineering

The final model was deliberately restricted to fields available in both the development/validation data and the December chart input. Features include pickup, delivery, distance, equipment, weight, route, date-derived features (month, day, day of week, day of year, week of year and day index), `distance_per_weight`, and a missing-weight indicator. Missing weight is imputed with the development-set median for the engineered ratio. LightGBM handles remaining numeric missing values natively.

5. Model

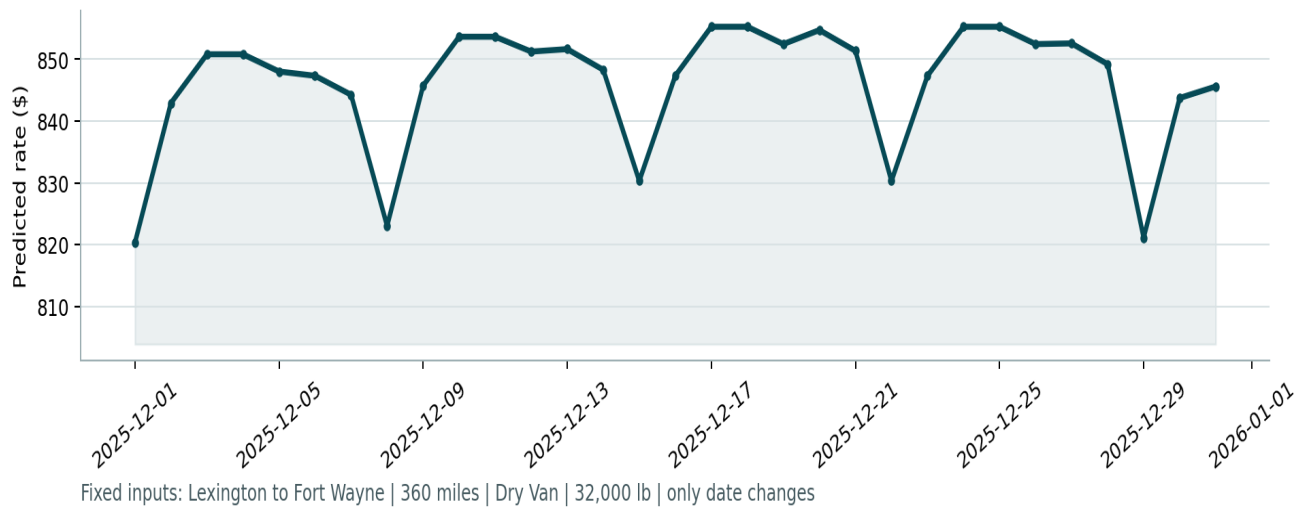
The selected model is LightGBM regression trained on $\log_{1p}(\text{posted_rate})$. The final configuration uses 300 estimators, `learning_rate=0.03`, `num_leaves=63`, `subsample=0.90`, `colsample_bytree=0.90`, L2 regularization of 1.0, and `random_state=42`. The log-target transformation reduced the effect of the highly skewed rate distribution and performed best in chronological validation.

6. Final Predictions

After validation, the model was retrained on all 48,000 labeled development rows. Predictions were generated for all 12,000 validation loads. A separate prediction was generated for each of the 31 December 2025 dates using the fixed December inputs supplied in the assessment.

7. Required December Prediction Chart

Candidate: December 2025 Predicted Load Rate



The supplied scorer was run successfully. It validated 12,000 final predictions and 31 December predictions and generated the chart above.

8. Submission Contents

The GitHub package contains the reproducible training/prediction script, requirements, README, validation_predictions.csv, December predictions, supplied scorer, generated chart, and this report.